dynamics information, but this may or may not be essential to the problem at hand: we often only wish to know the average of a given function $\phi(\boldsymbol{q})$ with respect the configurational density $\exp(-\beta U(\boldsymbol{q}))$, where U is the potential energy function of a molecular system. Nevertheless, Metropolis Monte-Carlo can encounter additional problems in the setting of molecular modelling. Where steep potentials (Lennard-Jones potentials, for example) and stiff bonds are present, Monte-Carlo methods may experience a high rejection rate. From another perspective, the "interesting region" for sampling a Gibbs distribution is an extremely small portion of the accessible phase space, since the measure of phase space is highly concentrated and localized by the inherent geometric constraints of the problem and/or steep potentials. As the dimensionality increases, the accessible phase space becomes thinner. A sampling algorithm must be guided from place to place in a large dimensional phase space riddled with steep barriers, and blind search typically leads to a lot of wasted effort. For this reason molecular dynamics, which benefits from acceleration information (the forces) to guide timestepping, and for which every step is accepted, is often superior to crude Monte-Carlo for molecular sampling. For a comparative discussion of Monte-Carlo and molecular dynamics methods, see [6, 278].

We hasten to add that there are many situations in which Monte-Carlo methods have the advantage. For example, we might have specific knowledge of a catalog of large scale structural "moves" that would allow our algorithm to scale energetic barriers without drastically biasing the sampling outcome. An example is when an assembly consisting of several component macromolecules is moved by rearranging the pieces, since it may take atomistic MD a very long time to make such moves. In those cases we would likely wish to make use of the system-specific knowledge that we have in order to accelerate the sampling. An alternative to assuming, a priori, a detailed book of moves, is to consider methods that adaptively construct good proposals, for example taking advantage of parallel computing methods. At the very least, one would expect that such methods need to take into consideration the local forces to avoid wasted steps. A practical way to do this is the Hybrid Monte-Carlo method mentioned in Appendix C.

Note that deterministic methods (e.g. Hamiltonian dynamics itself) can sometimes be used for molecular sampling, but without stochastic perturbation, such methods are not rigorously ergodic, since the Liouvillian operator does not possess a spectral gap and does not provide a unique stationary solution. The stochastic methods of this chapter are general purpose techniques that require little in the way of system-specific knowledge. They are workhorse methods which provide rigorous control of the canonical distribution. The challenge in using Langevin dynamics is to design maximally efficient methods that provide accurate averages for a wide range of observables using as large stepsizes as possible, and which converge as rapidly as possible. It is possible—even very easy—to build schemes that are wasteful, for example subject to unnecessarily stringent numerical stability barriers, and eliminating such mistakes is essential. This can be achieved by careful attention to the principles by which integrators are designed. Well designed stochastic differential equation approaches introduce dissipation without destroying